

INTERACTIVE APPLICATION GUIDE

The Complete System

Walkthrough

A step-by-step companion guide designed for clients, partners, and stakeholders to test and explore CycleTrace alongside a live browser window.

APPLICATION VERSION

CycleTrace 1.0 (Production)

BACKEND ARCHITECTURE

Appwrite Cloud & PayFast

PUBLISHED FOR

Client Review & Stakeholder Walkthrough

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How to Use This Manual

This manual is structured so that a client or reviewer can place this document side-by-side with an active browser window. Every subsection contains a highlighted **SIDE-BY-SIDE ACTION** box with exact keystrokes, button names, test input values, and expected UI updates.

Desktop Browser Setup

Open Google Chrome, Safari, or Edge to `http://localhost:3000` or your live deployment URL. Keep developer tools console open if inspecting network requests.

Live Appwrite Integration

CycleTrace connects to Appwrite Cloud with live session cookies, table storage, and secure photo buckets. No dummy data is used—actions persist immediately.

System Architecture & Security

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CycleTrace is built with a verification-first approach designed specifically for South African bicycle ownership realities. It replaces fragmented paper records and social media theft posts with an immutable registry database.

Technical Foundation

LAYER	TECHNOLOGY	FUNCTIONAL PURPOSE
Frontend	Next.js 15 (App Router), React 19, TypeScript	Server components, streaming layouts, and responsive client interfaces.
Authentication	Appwrite Auth & Session Manager	Secure email/password authentication with automatic session cookies and role preferences.
Database	Appwrite TablesDB	Stores bike records, transfer tokens, organization applications, and member rosters.
Storage	Appwrite Storage Bucket	Encrypted storage for bike photographs with strict role-based read/update permissions.
Onboarding	OnboardJS React Engine	Interactive product tour with local storage persistence and Appwrite preference syncing.
Compliance	POPIA Compliance Layer	Data minimisation, public serial verification without exposing personal contact details.

Security & Privacy Principles

- **Public Anonymity:** Anyone can verify a serial number publicly, but private owner details (full name, phone, residential address) are strictly protected and never displayed on public pages.
- **Role-Based Access:** Only bike owners can initiate transfers or flag thefts. Insurers and store owners must undergo admin verification before accessing elevated organizational workspaces.
- **Transfer Token Integrity:** Ownership transfers utilize cryptographic unique tokens with designated recipient email binding to eliminate fraudulent reassignment.

Public Portal & Serial Search

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The public portal enables potential second-hand buyers, bike shops, and law enforcement to instantly verify a bike's ownership and theft status before money changes hands.

STEP 1 Landing Page Overview & Live Search Bar

Navigate to [/](#). The homepage displays CycleTrace's mission, how it works, testimonials, and a central serial search input.

SIDE-BY-SIDE ACTION:

Scroll to the search input under "Before you buy, check the trace". Enter a brand, model, or known serial number (e.g. [Specialized](#) or [WSBC](#)) and press Enter or click

[SEARCH REGISTRY →](#)

STEP 2 Live Registry Search Results ([/search](#))

The registry query executes against the live Appwrite database. Matching bicycles are displayed with their photo, brand, model, color, registration year, and current protection status.

SIDE-BY-SIDE ACTION:

On [/search](#), click on any bicycle card result to open its detailed public record at [/bikes/\[id\]](#).

STEP 3 Public Bike Record Verification ([/bikes/\[id\]](#))

This page serves as proof of registry:

- **Protected Status:** Displays a green checkmark indicating the bike is legitimately registered and clear of theft reports.
- **Stolen Alert:** If flagged by the owner, a prominent red alert banner displays: *"This bike has been reported stolen. Do not purchase it."*
- **Frame Verification:** Highlights the registered serial number so a buyer can inspect the physical bike frame on the spot.

Authentication & Smart Redirections

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CycleTrace features unified authentication state management. The navigation bar dynamically adapts to user state, and guest auth pages intelligently redirect signed-in users.

Authentication State Matrix

PAGE / LOCATION	SIGNED-OUT VISITOR	SIGNED-IN MEMBER
Homepage Navbar (Desktop)	Shows "Log in" and "Register a bike" button	Shows "Dashboard" link and "Sign out" button
Homepage Navbar (Mobile Drawer)	Shows "Log in" and "Register a bike"	Shows "Dashboard" link and "Sign out" action
DemoShell Topbar (Public)	Shows "Log in" and "Get started"	Shows "Dashboard" and "Sign out" button
DemoShell Sidebar (Workspace)	N/A (Workspace requires login)	Displays User Avatar, Name, Email, and Sign-out button (<code>FiLogOut</code>)
Login Route (<code>/login</code>)	Displays member access email/password form	Automatically redirects to <code>/dashboard</code>
Register Route (<code>/register</code>)	Displays 3-step registration flow	Automatically redirects to <code>/dashboard</code> (Unless in <code>?mode=bike</code>)

STEP 1 Testing Sign-Out & Sign-In

SIDE-BY-SIDE ACTION:

1. Click **SIGN OUT** from the topbar or sidebar.
2. Notice how the navbar immediately changes back to "Log in" and "Register a bike".
3. Navigate to `/login`. Enter your credentials and click **SIGN IN →**.
4. Observe immediate redirect to `/dashboard` with the topbar and sidebar personalized to your account.

STEP 2 Verifying Guest Page Protection

SIDE-BY-SIDE ACTION:

While logged in, manually type `/login` or `/register` in your browser address bar and hit enter. The application immediately detects your active session and redirects you to `/dashboard` without flashing the forms.

Rider Dashboard & Walkthrough Tour

The Rider Dashboard (/dashboard) is the central command center for bicycle owners to oversee their registered fleet, inspect live records, and trigger recovery actions.

Interactive Onboarding Walkthrough (OnboardJS)

First-time users entering the dashboard are greeted with an interactive 3–step onboarding walkthrough guide:

STEP 1

Welcome to CycleTrace

Introduces the workspace where riders protect bikes and manage records.

STEP 2

Your Bikes Live Here

Points to the "My bikes" grid, tied to your secure Appwrite account ID.

STEP 3

Know Your Next Action

Shows how to register new bikes, transfer ownership, or flag theft.

SIDE-BY-SIDE ACTION:

1. Advance through the tour steps using the green

NEXT →

button.

2. Test the

BACK

button to return to a previous step.

3. Click

FINISH ✓

on the final step.

4. Notice the tour closes cleanly, and a floating

TOUR GUIDE

button appears in the bottom-right corner.

5. Click

TOUR GUIDE

to restart and replay the tour at any time.

Dashboard Key Metrics

- **Protected Bikes:** Total number of bikes registered under your Appwrite account currently in good standing.
- **Reported Stolen:** Real-time count of bikes actively flagged as stolen.
- **Account Status:** Confirms live active Appwrite session connectivity.

Adding Additional Bikes to Your Fleet

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When a logged-in rider wants to register another bicycle, they bypass account creation and jump directly to bicycle details.

STEP 1 Initiate Bike Registration (/register?mode=bike)

From the dashboard, click **Register another bike** or the sidebar/topbar **Add a bike** button.

SIDE-BY-SIDE ACTION:

Notice that Step 1 ("Account") is automatically marked ready. The screen starts directly on

01 BIKE DETAILS

STEP 2 Enter Bicycle Identifying Details

FIELD	EXAMPLE INPUT	VERIFICATION REQUIREMENT
Brand	Trek, Specialized, Cannondale, Giant	Exact manufacturer name.
Model	Domane SL 6, Stumpjumper, Synapse	Model and specification grade.
Year	2024	Model manufacturing year.
Colour	Matte Black / Hyper Yellow	Distinctive frame paint finish.
Serial Number	WTU12345678Z	Stamped on underside of bottom bracket.
Storage Location	Cape Town / Gardens	Primary residential city or suburb.
Photo Upload	Side profile image (< 10MB)	Clear picture showing frame geometry.

STEP 3 Review & Submit to Appwrite

Click **Continue** → to inspect the summary card. Confirm ownership accuracy checkbox, then click **Complete registration**.

EXPECTED OUTCOME:

A success confirmation screen displays your new unique **REGISTRATION ID** with links to view the public record or return to your dashboard.

Theft Reporting & Recovery Workflow

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When a bicycle is stolen, every hour counts. CycleTrace allows immediate status escalation so any buyer or bike shop checking the serial number is warned.

STEP 1 Access Theft Reporting (/dashboard/report)

Navigate to `/dashboard/report` from your dashboard action menu.

SIDE-BY-SIDE ACTION:

1. Select the bicycle from the dropdown list of your owned bikes.
2. Enter the date of incident, last seen location, and police case number (SAPS CAS).
3. Add any distinguishing marks (e.g. scratched top tube, aftermarket pedals).
4. Click **FLAG AS STOLEN →**

STEP 2 Verify Public Registry Escalation

CycleTrace updates the bike's record status in Appwrite from `protected` to `stolen` instantaneously.

SIDE-BY-SIDE ACTION:

Open an incognito or private browser tab. Go to `/search` and search for the serial number of the bike you just flagged. Observe the high-visibility red **REPORTED STOLEN** alert banner on its public page.

COMMUNITY PROTECTION MECHANISM:

The public page includes a direct reporting action for anyone spotting the bike to securely submit tips to `reports@cycletrace.co.za` without exposing the owner's personal details.

Secure Ownership Transfer Protocol

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Selling or giving away a bicycle? CycleTrace enables verifiable cryptographic handovers, updating the provenance record and transferring management rights safely.

Two-Party Transfer Workflow

SENDER FLOW Initiating Transfer (/dashboard/transfer)

SIDE-BY-SIDE ACTION:

1. On `/dashboard`, locate the bike card and click **TRANSFER OWNERSHIP**.
2. Enter the buyer's email address (e.g. `buyer@example.com`).
3. Agree to release ownership rights and click **GENERATE TRANSFER LINK →**.
4. Copy the unique transfer URL containing the secure handover token.

BUYER FLOW Accepting Ownership (/transfer/[token])

SIDE-BY-SIDE ACTION:

1. Paste the transfer URL in a browser.
2. Review the bike details on the transfer reception page.
3. Enter the buyer email to verify authorization.
4. Click **ACCEPT OWNERSHIP →**.

POST-HANDOVER STATE:

The bike is immediately removed from the seller's active fleet and assigned to the buyer's account. The public provenance history logs the transfer timestamp.

Organizations & Commercial Roles

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CycleTrace partners with bike shops, insurers, and recovery organizations. The Onboarding Hub ([/onboarding](#)) routes commercial teams to the appropriate workflow.

Store Owner Workspace ([/store-owner](#))

Allows bicycle retail stores to register bicycles for customers directly at the point of sale. Ensures new owners leave the shop with verified digital protection active on day one.

- Batch customer bike registrations
- Custom store branding on handover
- Proof of authorized retail chain

Insurer Workspace ([/insurer](#))

Gives underwriting and claims teams instant serial verification tools. Insurers verify frame numbers before writing policies and validate claims history.

- Instant serial fraud & duplicate checks
- Active claims queue review
- Confidence verification scores

Store Owner & Insurer Application Flow

ACTION Submitting Organization Verification

SIDE-BY-SIDE ACTION:

1. Navigate to [/onboarding](#) .
2. Click

STORE OWNER

(or

INSURER

).

3. Fill in the business registration number, contact person, phone, and address.

4. Click

SUBMIT APPLICATION →

. The submission is queued in Appwrite under [organization_applications](#) for platform admin review.

Platform Administration & Governance

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The Platform Administration portal (`/admin`) allows authorized operators to review organization applications, manage memberships, and maintain registry integrity.

Admin Capabilities

MODULE	FUNCTIONALITY	ADMINISTRATIVE CONTROLS
Application Queue	Review pending retailer and insurer registrations	Approve application, reject with notes, or request documentation.
Organization Roster	Active verified partner organizations	Suspend access, invite team members, adjust role quotas.
Registry Analytics	Platform-wide bike registration health	Inspect total registered bikes, active theft rates, and transfer volume.

SECURITY

Admin Access Control

Admin routes are strictly guarded by Appwrite labels (`admin`) or the `ADMIN_EMAILS` environment allowlist. Unauthenticated or unauthorized visitors are automatically blocked.

Settings, Profile & Billing

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Members manage their personal details, Appwrite session security, and subscription billing via [/settings](#).

Account Preferences ([/settings](#))

- **Full Name:** Updates user display name across ownership records and certificates.
- **Email Address:** Read-only display of primary Appwrite account identity.
- **Contact Phone:** Stored securely for recovery notifications.
- **Home City:** Primary locality for regional recovery network alerts.
- **Security:** Direct visibility of active Appwrite Auth session encryption.

Billing & Subscription ([/settings/billing](#))

CycleTrace integrates with South Africa's leading payment gateway, **PayFast**, to manage monthly bike registry subscriptions (R24.99 / month).

PRODUCTION BILLING NOTE:

On staging and evaluation environments without live merchant keys, the billing portal displays a clean, informational state indicating active Appwrite records are maintained without service interruption.

Interactive Testing Checklist

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Use this verification matrix to conduct a complete evaluation of the CycleTrace application. Check off each scenario as you complete the walkthrough.

✓	TEST SCENARIO	VERIFICATION STEPS	EXPECTED RESULT
[]	Public Search	Go to <code>/search</code> , query a bike serial number or brand.	Matching bikes appear with photo and protected status.
[]	Public Record View	Click a bike result card to view <code>/bikes/[id]</code> .	Verified badge and frame specs displayed; owner data remains private.
[]	New User Register	From homepage, click Register a bike while signed out.	Step 1 creates account, Step 2 captures bike, Step 3 confirms.
[]	Auth Redirection	While signed in, navigate to <code>/login</code> or <code>/register</code> .	Instantly redirected to <code>/dashboard</code> .
[]	Navbar Auth Sync	Check homepage and topbar after signing in and signing out.	Changes between "Log in / Register" and "Dashboard / Sign out".
[]	Onboarding Tour	Open <code>/dashboard</code> . Step through the 3-card guide.	Cards animate smoothly; "Tour guide" button replays on demand.
[]	Add Another Bike	From dashboard, click Register another bike .	Starts directly on Bike Details; no password required.
[]	Flag Theft	Go to <code>/dashboard/report</code> and submit a theft flag.	Status changes to stolen; public search shows red alert banner.
[]	Ownership Transfer	Generate transfer token on <code>/dashboard/transfer</code> .	Recipient accepts on <code>/transfer/[token]</code> ; record reassigns.
[]	Commercial Hub	Visit <code>/onboarding</code> and select Store or Insurer.	Application forms submit cleanly to Appwrite admin queue.
[]	Profile Settings	Update phone and city on <code>/settings</code> and save.	Preferences save directly to Appwrite with green success note.

Frequently Asked Questions & Support

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Common Questions

Where do I find my bike's serial number?

On 95% of bicycles, the serial number is stamped into the metal on the underside of the bottom bracket shell (beneath the pedals). It can also sometimes be found on the rear dropout or the head tube.

Can someone steal my bike by claiming my serial number?

No. Once a serial number is registered in CycleTrace, any attempt to register the same serial number triggers an automatic **409 Conflict** guard. Ownership can only be updated through an authorized cryptographic handover link from the existing registered owner.

What if I sell my bike to someone who isn't on CycleTrace yet?

Simply generate a transfer link using their email address. When they open the link, CycleTrace allows them to accept ownership and create their free secure account in one smooth step.

How does CycleTrace protect owner privacy?

In accordance with the South African Protection of Personal Information Act (POPIA), public search results only disclose bicycle technical specifications and theft verification status. Your full name, residential address, and contact numbers are strictly confidential.

Technical Support & Enquiries

For technical inquiries, partner onboarding, or pilot deployment support, please contact the CycleTrace team:

- **General Enquiries:** hello@cycletrace.co.za
- **Theft Recovery & Law Enforcement:** reports@cycletrace.co.za
- **Privacy & Compliance:** privacy@cycletrace.co.za
- **Official Website:** <https://cycletrace.co.za>